INTELLIGENT TRANSPORTATION SYSTEMS

Dynamic Traffic Monitoring & Alternate Path Routing

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PART 01

System Foundations

Modeling urban road networks as dynamic, time-dependent graphs with live telemetry ingestion.

Graph Network Modeling

- **Vertices (V):** Intersections, highway junctions, ramps, and telemetry checkpoints across the city map.
- **Directed Edges (E):** Road segments connecting intersections, accounting for one-way streets and multi-lane turn restrictions.
- **Dynamic Edge Weights:** Real-time travel time function incorporating speed, congestion density, and signal delays.
- **IoT Telemetry:** High-frequency updates from GPS probes, loop detectors, and traffic cameras constantly modifying weight parameters.

Data Structure Selection

Adjacency List with Dynamic Edge Hashes

Structure: Graph represented as array of adjacency lists supplemented by an edge lookup hash map.

Justification: Road networks are sparse graphs where average vertex degree . Adjacency matrix consumes space, whereas Adjacency List requires only space complexity, optimizing memory footprint for millions of nodes.

Min-Priority Queue (Fibonacci / Indexed Heap)

Structure: Priority Queue supporting key updates ( ) during shortest path frontier expansion.

Justification: A Fibonacci Heap delivers amortized time for and for , accelerating path search frontier updates under rapid traffic flow changes.

Core Real-Time Operations

1. Dynamic Weight Update

When congestion occurs on edge , the weight is updated in real time via direct hash indexing in worst-case time.

2. Point-to-Point Query

Calculating fastest travel route from source to target using dynamic shortest path search optimized by bidirectional heuristics.

3. Alternative Path Discovery

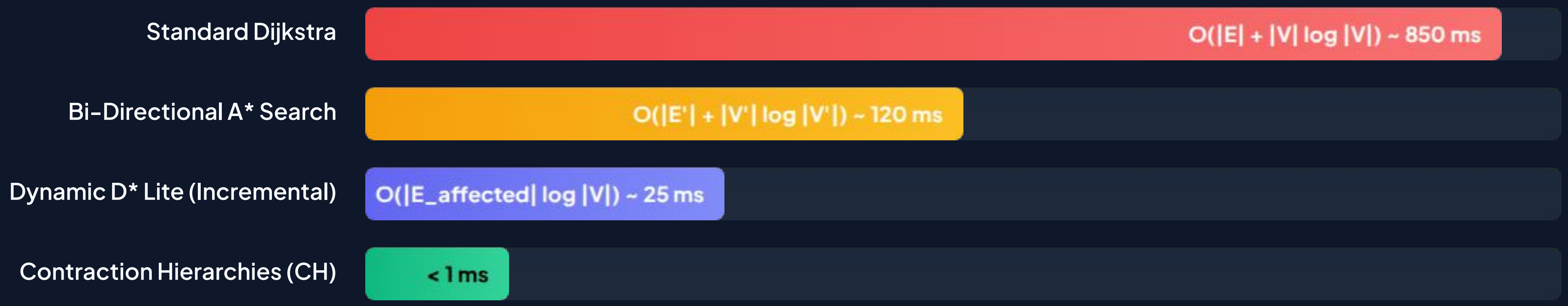
Finding -shortest bounded-cost distinct paths (Plateau / Penalty methods) to avoid rerouting all drivers onto a single road segment.

PART 02

Algorithmic Analysis

Evaluating search algorithms and dynamic graph structures under complexity constraints.

Pathfinding Algorithm Benchmarks



Comparison of query execution latency on a standard city road graph (|V| = 1,000,000 vertices). Contraction Hierarchies and Dynamic D Lite significantly outperform static search algorithms.*

Dynamic Congestion Event

- **Incident Detection:** Sensors detect speed drops, triggering a weight change on segment .
- **Local Re-Search:** Incremental algorithms (D* Lite) recompute affected search trees in time without global reset. $O(|E_{\text{affected}}| \log |V|)$
- **Alternate Dispersal:** Generates disjoint paths to prevent bottlenecking secondary arteries.

Contraction Hierarchy Speedup

<1 ms

Query Time on 1M+ Node Map

Sub-Millisecond Alternate Path Lookups
Contraction Hierarchies (CH) order vertices by importance and add "shortcut" edges to bypass less critical nodes. During runtime, a bidirectional Dijkstra search operates strictly on the upward hierarchy.

This reduces search space from millions of nodes to fewer than 500 settled nodes, enabling centralized servers to serve tens of thousands of dynamic vehicle rerouting queries per second.

Alternate Routing Paradigm

Bounded Cost Alternatives

Filters alternate paths where total travel time .

Flow Load Balancing

Distributes traffic across disjoint paths to avoid creating new secondary bottlenecks.

Connected V2X Guidance

Pushes dynamic updates to vehicles via telemetry endpoints with low latency response.

Questions & Answers

Thank you for your attention

Intelligent Traffic Monitoring & Routing Architecture